Chirality and Agentic AI

Useful analogies, not safety proofs

Audience	A technically curious reader exploring chemistry-inspired AI controls; basic agent/tool concepts assumed, chemistry explained below.
Purpose	Separate factual chemistry, useful engineering intuitions, and claims requiring experiments.
Scope	Chemical mechanisms, candidate agent controls, preliminary software findings, and swarm-inspired research hypotheses; no claim of validated effectiveness.

Bottom line: stop unsupported intermediate claims from acquiring authority through repetition. Four candidate controls—evidence-seeking criticism, claim quarantine, pre-action checks, and evidence-preserving restarts—offer a research direction, but their effectiveness cannot be derived from molecular chemistry.

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1 What the chemistry actually establishes

Chirality is molecular handedness: enantiomers are non-superimposable mirror-image forms, conventionally designated R and S where applicable. Their relative amounts can be summarized by the magnitude of enantiomeric excess, $\text{ee} = \frac{|n_R - n_S|}{n_R + n_S}$, for a nonempty two-enantiomer mixture; this composition statistic says nothing about which outcome is desirable. [1]

1.1 Nobel context and chemical mechanisms

2026 Nobel Prize: Henri B. Kagan and Kenso Soai received the chemistry prize "for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis," announced on 7 October 2026. **2021 Nobel Prize:** Benjamin List and David W. C. MacMillan received the chemistry prize "for the development of asymmetric organocatalysis"; these are related contributions, not the same award or mechanism. [2], [3]

Frank's 1953 model combines self-reproduction with mutual antagonism: opposite enantiomers deactivate one another. Crucially, the combination can **amplify an initial imbalance**, rather than restore balance; it does not chemically recognize a "wrong" handedness, so symmetric opposition is not inherently a stabilizing mechanism. [1]

Represent the deactivation schematically as $R + S \rightarrow I$, with I an inactive heterochiral species. Calling every such species a **meso compound** is unjustified without its structure: a mixed-handed aggregate is not automatically a molecule with the internal symmetry required for meso classification.

Kagan-type nonlinear effects mean that product ee need not vary linearly with catalyst ee. In reservoir models, heterochiral aggregates sequester both catalyst enantiomers and can enrich the remaining active fraction; the relevant aggregates are not generically "bad" catalyst molecules, and their activities and equilibria matter. [1], [4]

Some models have catalytically inactive heterochiral species; others have less-active heterochiral species that produce racemic product. Such sequestration can **enhance** stereochemical bias—another reason it does not prove that memory quarantine suppresses an agent's confirmation bias. [1]

Soai's reaction demonstrates asymmetric autocatalysis with amplification of enantiomeric excess: the product-derived zinc alkoxide catalyzes further formation of the same product. Soai and colleagues reported the key discovery in 1995; product/catalyst aggregation, substrate binding, and relative species activities are essential to understanding how this feedback operates. [5], [6]

Use simple dimer pictures as explanatory models, not a settled universal structure for Soai chemistry: Athavale et al.'s 2020 experimental/computational study develops a tetrameric catalyst and transition-state model. Demonstrating that a laboratory reaction can generate very high ee also does not establish that this reaction actually caused biological homochirality on early Earth. [1], [6]

Racemization is interconversion that reduces enantiomeric enrichment toward a racemate under suitable conditions. A 1:1 molecular equilibrium is the idealized expectation for enantiomers interconverting in an achiral environment without selective removal; heating alone does not guarantee rapid, clean racemization, and coupling interconversion to phase separation can give different overall outcomes. [1]

Asymmetric organocatalysis favors one stereochemical pathway using a chiral organic catalyst. Favoring a pathway is not the same as making every competing pathway impossible; electronic activation and catalyst–substrate organization matter alongside steric effects. [7]

For two competing, kinetically controlled pathways with comparable prefactors, define $\Delta\Delta G^{\ddagger} = G^{\ddagger}_{\text{minor}} - G^{\ddagger}_{\text{major}}$. The usual transition-state estimate gives

$$\frac{k_{\text{major}}}{k_{\text{minor}}} \approx \exp\left(\frac{\Delta\Delta G^{\ddagger}}{RT}\right).$$

Here R is the gas constant and T the absolute temperature; a finite barrier difference gives finite selectivity, not a guarantee of zero unwanted product. This relationship is chemical background, consistent with transition-state/selectivity comparisons in the mechanistic literature, not a model of how a validator changes an LLM's error probability. [6]

2 Four analogies, with their limits visible

Chemical inspiration	Proposed AI control	Boundary of the analogy
Competing enantiomers	Evidence-seeking critic	Opposed claims do not cancel like molecular species; multiple explanations can both contribute.
Catalyst sequestration	Quarantine unverified claims	A prompt label does not physically isolate information or establish truth.
Stereoselective catalysis	Pre-execution gateway	A schema enforces structure; permissions and semantic checks enforce only their stated rules.
Racemization	Restart a stagnant search	Do not restore equal belief in unequal hypotheses or erase evidence and constraints.

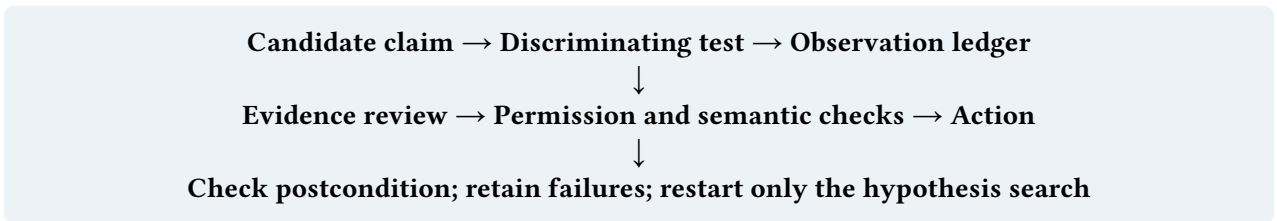


Figure 1: Proposed control flow, not an experimentally validated architecture. Claims require observations before they guide consequential action; failed checks preserve evidence and return to testing.

2.1 Evidence-seeking critic, not an agent paid to disagree

Principle: challenge a favored explanation before acting on it. Neither a mirror-image prompt nor a second model establishes statistical independence or creates evidence by cancellation.

Instead of asking one agent to "prove the database is broken" and another to "prove it is normal," ask both to compare database, network, application, and unknown-cause hypotheses against the same incident. Let the critic identify the strongest disconfirming observation and propose a discriminating test; allow mixed causes and abstention.

Multiagent debate has reported improvements on selected reasoning and factuality tasks, but that does not validate this particular controller. Intrinsic self-correction without external feedback sometimes degraded performance in the reasoning experiments of Huang et al.; neither result is a universal theorem about current agents. [8], [9]

Limits of subtraction: a rule such as $A'_t = \max(0, A_t - \alpha B_t)$ is not a consequence of Frank kinetics. Without a defined evidence scale and calibrated α, it treats correlated statements as additive and can suppress a correct minority hypothesis; $B_t \approx A_t$ is not guaranteed.

Practice: record observations for and against each explanation, identify shared sources, and choose the next test by its ability to distinguish explanations. A second opinion is useful when it supplies a different check, not merely a different conclusion.

2.2 Epistemic quarantine with provenance

Principle: prevent a tentative claim from becoming an established premise merely through repetition. An **evidence ledger** separates hypotheses, observations, and assessments without assuming that catalyst sequestration has a direct computational equivalent.

Record	What it contains
Hypothesis	A concise, revisable claim; proposed test; unresolved alternatives.
Observation	Source or tool, time, exact scope, result, and known reliability limits.
Assessment	Which observations support the claim, contradictions, uncertainty, and review date.

Promotion is claim-specific. A successful API request proves neither that its payload is true nor that it answers the intended question: a healthy database replica does not establish that the primary is healthy. Retrieved content can also contain indirect prompt injection, so external observations must remain data, not instructions or authorization. [10]

Use short claim summaries and tool-visible evidence, not a requirement to extract or persist private chain-of-thought. A textual `epistemic_status="speculative"` label is a convention, not an access-control mechanism; actual exclusion requires controlling retrieval and prompt assembly, and arbitrary attention masking is not a standard assumption for hosted APIs.

ReAct already interleaves reasoning, action, and external observations, and its paper reports reduced hallucination/error propagation on the evaluated question-answering tasks. Reflexion uses feedback and an episodic memory buffer, so the existing literature supports studying feedback-aware memory—not declaring all model-produced summaries harmful. [11], [12]

Limits: quarantining text does not establish a change of dynamical order from $k > 1$ to $k \leq 1$, and unsupported hypotheses do not automatically "decay." Define expiration or review rules explicitly, while retaining contradictory and historical evidence.

2.3 Pre-execution checks with asymmetric permissions

Principle: validate before effects occur, with stronger checks for more consequential actions. The useful asymmetry is an explicit permission and verification policy, not a chemical activation energy.

Require a structured action proposal containing the target, intended effect, supporting observation IDs, expected postcondition, a disconfirming condition, and a recovery plan where feasible. Enforce schema and semantic checks in the tool gateway outside the proposing model; a well-written justification is not authorization.

JSON Schema can constrain fields, types, enumerations, and additional properties, but structural validity does not establish truth or safety. A perfectly valid deletion request can target the wrong resource; unit tests cover their assertions, not all real-world outcomes. [13]

Risk is broader than read versus write. Reads can expose secrets, spend money, or invoke unsafe endpoints; writes may be transactional and reversible, and a Git commit is not inherently irreversible. Use least privilege, scope checks, freshness checks immediately before execution, and independent verification proportional to impact; a small checking model remains fallible.

An effectively infinite barrier exists only for a forbidden action that an unavoidable gateway actually rejects. It does not mean all bad actions are recognized, nor does it protect alternate execution paths.

2.4 Evidence-preserving restart, not erasure

Principle: stop an unproductive trajectory and reconsider alternatives. Restart from a neutral incident summary while preserving user constraints, observations, failed tests, permissions, and an append-only action history; pruning a search branch must not destroy the audit record.

The proposed cognitive enantiomeric excess,

$$\mathrm{CEE}_t = \frac{|p_1 - p_2|}{p_1 + p_2},$$

is merely a normalized two-branch imbalance when $p_1 + p_2 > 0$. It does not establish correctness, neglects other hypotheses, and requires probabilities with a defined interpretation; confidence expressed in prose is not a calibrated posterior.

If $p_1 + p_2 = 1$ and $p_1 \geq p_2$, then $\mathrm{CEE} = 2p_1 - 1$, so a threshold of 0.85 corresponds to $p_1 = 0.925$. Such a threshold, a three-attempt rule, or restarting three branches would be heuristic choices, not policies validated by the cited literature.

Monitor observable stagnation instead: repeated equivalent actions, unchanged test results, and no new relevant observations. If the proposal invokes **information gain**, specify a belief model and measurement; a count of new observations is only a proxy, not Shannon information gain.

Tree of Thoughts provides precedent for search, backtracking, and evaluated alternatives on specific tasks, not for this CEE-triggered policy. Different prompts or temperatures may diversify outputs but do not guarantee independent errors; raising temperature can also worsen reliability. [14]

2.5 Limits of mathematical and physical analogies

Not a law of ReAct: $P_{t+1} \propto P_t^k$ lacks a normalization or update rule. For $0 < P_t < 1$ and $k > 1$, $P_t^k < P_t$; even the simplest normalized binary update amplifies the initially larger branch, not necessarily the incorrect branch.

For illustration only, if $p' = \frac{p^k}{p^k + (1-p)^k}$, then $p = 0.2$ and $k = 2$ give $p' \approx 0.059$, while $p = 0.8$ gives $p' \approx 0.941$. Even this simple update can increase or decrease an error probability depending on the starting point; it is not an empirical model of an agent.

Not quantum noise: nonzero sampling temperature changes a computational sampling distribution; it does not establish a quantum mechanism, and an error is not inevitable in every sampled answer. Low or zero temperature is also no guarantee of truth.

3 How to test the ideas rather than just rename them

Objects and measurements: chemical concentrations and enantiomeric excess belong to a molecular reaction; agent claims, evidence records, actions, and task outcomes belong to a software process. The research question is whether the proposed software controls improve these measured outcomes, not whether their terminology resembles chemistry.

Candidate	Competing hypothesis	Falsifier
A · Conservative	A simple evidence ledger plus deterministic action checks supplies most of the benefit.	Matched-budget trials show no reduction in unsupported claims or unsafe attempts versus ordinary tool-grounded operation.
B · Analogical	An evidence-seeking critic adds benefit beyond ledger and gateway checks.	A compute-matched extra test or independent solve performs as well or better; debate adds delay or false rejection.
C · High-risk	A stagnation-triggered restart improves recovery without losing constraints.	Restarts repeat failed actions, erase essential evidence, increase violations, or provide no matched-budget recovery gain.

Protocol: use sandboxed debugging and fact-checking tasks with hidden ground truth, including database, network, application, mixed, and unknown causes. Randomize a misleading early hint, a correct hint, or no hint; include stale or misleading tool output and conflicting observations, without exposing real services or secrets.

Comparators: evaluate a tool-grounded baseline, each of the four controls separately, and the combined system. Match total token/tool budgets; include a sham critic that merely paraphrases and an equal-budget additional-observation condition to distinguish useful evidence from extra computation.

Outcomes: predeclare task success, unsupported-claim promotion rate, unsafe action attempts, executed violations, false blocks, recovery rate, cost, and latency. Use repeated seeds and task families, paired comparisons and uncertainty intervals; report model versions, prompts, gateway policy, sample size rationale, and all exclusions.

Kill criteria: reject the claimed benefit if improvement disappears under budget matching or if safety gains rely on refusing nearly every action. Stop the combined proposal if it raises executed violations or loses required constraints; tune any thresholds on development tasks and evaluate once on held-out tasks.

Feasibility: the proposed controlled evaluation is not model training; start with simulated tools and a fixed call/spend cap. The preliminary trials below do not implement this full protocol, so effectiveness, novelty, and any chemical-to-computational causal equivalence remain unverified.

4 Preliminary software findings

Status · 19 October 2026: two developmental exercises used Claude Sonnet 4.6, high variant, through GitHub Copilot. Their value is identifying failure modes and better experiments—not demonstrating that chemistry-inspired controls improve autonomous agents.

4.1 Prompt-level smoke test

Four fresh subagent sessions each handled the same four synthetic cases: misleading database evidence, a report with unwanted metadata and an injected instruction, repeated unsuccessful pool-size changes, and a clear reversible spelling correction. Conditions were ordinary guidance, an observation/inference ledger, a targeted countercheck, and a constraint-preserving reset prompt; these were prompt overlays, not independently enforced components.

Condition	Criteria met / 11	Cases passed / 4
Ordinary guidance	9	2
Evidence ledger	9	2
Targeted countercheck	10	3
Recovery/reset prompt	9	2

Scoring sensitivity: a case passed only when all its criteria passed. The countercheck result accepts a later explicit qualification despite earlier categorical wording; rejecting that contradiction leaves every condition at 2/4 cases. A more permissive target-label-only interpretation yields 2/3/4/2 cases respectively, so the apparent ranking is not robust.

Observed strengths: all sessions rejected the injected instruction, avoided recommending another identical failed intervention, and approved the simulated spelling correction without claiming to have applied it. All actual tools were read-only; this cannot establish safety at a real mutation boundary.

Persistent failures: replica reachability was overextended into conclusions about the primary database or a specific DNS fault. Low pool-utilization observations were promoted into broad exclusions of pool exhaustion; a correct proposed action did not guarantee a correctly bounded explanation.

Session findings: every condition read six fixture files without repetition, but baseline and reset also listed a directory outside the file-only boundary. All four dispatched task-source reads before role restoration completed; three also executed them early. Final responses grew from 415 words at baseline to 520, 594, and 644 words, without a convincing quality gain.

Limits: one unblinded session per condition, fixed case order, shared-model errors, existing baseline safeguards, subjective grading, and unmatched compute prevent efficacy estimates. This was not a test of a separate independent critic, a runtime gateway, or a validated stagnation detector.

4.2 Direct instruction integration

A second exercise changed existing evidence, investigation, memory-checkpoint, and dispatch guidance rather than adding another prompt overlay. Two local Python/SQLite episodes each used a fresh diagnosis producer and a fresh repair consumer: baseline before the changes, revised after them; original tests, measurement data, and task instructions were preserved.

Code result: both repairs changed writes from the replica to the primary, preserved duplicate-order errors, and moved the original suite from two failures and one pass to three passes. Both also corrected the README spelling error; successful code tests did not verify the accompanying report.

First-pass failure: both reports still claimed "Pool exhaustion has been ruled out for the sampled observation window" from only three measurement instants. All four sessions skipped supporting-reference reads, and three omitted the requested memory-skill activation, even though revised workers received the updated investigation-skill body.

Assisted correction: after explicit feedback and named prerequisite paths, the same revised repair session activated memory, loaded four supporting references, and narrowed the report to "Pool exhaustion was not observed in the sampled measurements." Only the report and completion record changed in this correction; it demonstrates assisted recovery, not spontaneous compliance or a causal effect of the instruction edits.

Handoff and protocol gaps: a baseline handoff labeled failed checks as absent despite recording two test failures; the revised historical handoff retained overbroad window language. Workers also used native editing interfaces rather than the initially requested patch-only interface; later permission to use those interfaces did not retroactively satisfy the original protocol.

Limits: this was one ordered pair with common evidence-scope priming and earlier outcome commentary visible to revised workers. There was no randomization, blinding, held-out evaluation, or exact budget matching. No MCP service, host-enforced gate, induced recovery loop, or production action was tested.

4.3 Implications for the next iteration

- **Bound the claim, not just its label:** record source, target, revision, conditions, and actual sampling coverage; independently assess whether the conclusion exceeds them. A healthy replica does not establish primary health, and three unsaturated samples do not cover the intervals between them.
- **Make prerequisites observable:** name required reference paths and check that loads complete before dependent work. A skill invocation or delivered body is not evidence that its supporting references were loaded; a completed read still does not establish comprehension.
- **Keep failures separate from recovery:** retain first-pass conclusions, failed checks, permissions, unknowns, and correction history. Passing application tests, correcting prose, and following the process are distinct outcomes.
- **Test one change at a time:** use fresh repeated sessions, counterbalanced assignment, frozen task context, held-out tasks, and matched budgets. Include correct-initial-hypothesis and easy-action controls; measure unsupported claims, false blocks, constraint retention, and overhead separately.

- **Escalate enforcement only for an evidenced need:** a host-side ordering or receipt validator could test prerequisite and scope metadata, but not prove natural-language entailment. An optional MCP check cannot intercept unrelated native tools; any enforcement claim needs explicit coverage and bypass tests.

Research conclusion: the supported finding is a gap between stated safeguards, actual loading, and inference restraint. No candidate has yet demonstrated robust autonomous benefit; the useful next target is observable prerequisite handling plus scope-bounded evidence, with failure-preserving evaluation rather than a more elaborate chemical metaphor.

5 Swarm intelligence: quorum without manufactured certainty

Research question: can collective evidence gathering, quorum-like commitment, and cross-inhibitory criticism improve the controls above without amplifying shared errors? The connection is mechanistic—feedback and commitment policies—not a claim that molecular reactions, insect colonies, and language-model teams are equivalent systems.

5.1 What quorum and criticism do differently

Sumpter and Pratt describe quorum responses as sharply nonlinear changes in an individual's behavior as the number of others already exhibiting a behavior increases. Recruitment and local assessments can coordinate collective nest selection, but a quorum threshold is not necessarily a numerical majority of the colony; their models also show that strong feedback can amplify early errors into collective incorrect choices. [15]

Seeley and colleagues report that honeybee scouts deliver inhibitory stop signals preferentially to dancers advertising rival nest sites. Their analytical model links this cross-inhibition to resolving deadlock over equal sites; selecting one of two equally good options is a coordination achievement, not evidence that one option was factually false. [16]

Mechanism	Possible agent analogue	Important distinction
Recruitment / positive feedback	Allocate more investigation to promising hypotheses.	Popularity can reinforce a shared mistake; it is not new evidence.
Quorum response	A declared threshold for promoting a provisional conclusion.	Commitment, majority voting, and truth are different properties.
Cross-inhibition	A targeted challenge can interrupt unsupported promotion.	A red team seeks counterevidence; a biological stop signal is not a logical refutation.
Collective commitment	Coordinate the next permitted action.	Agreement cannot override authorization, a failed postcondition, or a safety constraint.

Connection to chirality: self-reinforcement and antagonism can select one branch in both chemical and collective-decision models. Neither mechanism identifies a desirable or correct branch by itself; the transferable research target is how evidence influences amplification and commitment, not whether opposition guarantees balance. [1], [15], [16]

5.2 Why five agreeing agents need not be five checks

For an idealized binary question, suppose an odd number n of voters have independent correctness events, each with the same probability p. Then majority correctness is

$$P_{\text{majority}} = \sum_{k=\frac{n+1}{2}}^{n} \binom{n}{k} p^k (1-p)^{n-k}.$$

With $p > \frac{1}{2}$, this idealized probability increases with odd group size; independence and competence are substantive assumptions, not consequences of creating more agents. Shared evidence and common causes can correlate errors even when voters do not communicate. [17]

Counterexample: five independent voters with $p = 0.7$ give majority correctness 0.83692. If all five instead copy one fallible judgment, their individual accuracy can remain 0.7 while majority accuracy is still only 0.7; unanimity adds no reliability in that case.

Implication: independent initial submissions can reduce direct copying, but different prompts, temperatures, roles, or model providers do not establish statistically independent errors. Track agent endorsements and distinct evidence origins separately: reanalyzing shared evidence may improve an inference, but it is not another observation of the environment. [17]

Multiagent debate has reported gains on selected reasoning and factuality tasks, providing a relevant precedent rather than a general consensus guarantee. The preliminary trials above did not test a swarm quorum policy or an independent debating group. [8]

5.3 Three competing, falsifiable directions

Direction	Causal hypothesis	Disconfirming result
A · Source-aware quorum	Independent initial assessments plus visible evidence lineage reduce repeated promotion of the same unsupported claim relative to a head-count vote.	The gain disappears against an equal-budget independent-solve baseline, or lineage rules merely increase false blocks.
B · Cross-inhibitory critic	An evidence-bearing challenge interrupts premature commitment better than passive voting.	A targeted extra observation performs as well or better, or criticism suppresses supported minority findings and correct initial hypotheses.
C · Adaptive recruitment	Additional workers are useful only when their assigned observations or analyses can change an unresolved decision.	A fixed small team achieves equal or better accuracy and constraint retention at equal cost; recruitment only accelerates agreement.

Candidate workflow, not an implemented architecture: collect initial assessments before sharing peer conclusions → expose observation sources and disagreements → request a discriminating check → apply a predeclared promotion rule → pass the proposed action through independent permission and postcondition checks. A verified constraint violation stops the affected action; mere dissent instead calls for assessment, not automatic rejection or forced unanimity.

Evaluation: compare one grounded agent, equal-budget independent solves with majority aggregation, ordinary debate, and each proposed addition separately. Hold evidence access and total budgets fixed within comparisons; separately vary shared versus distinct evidence and peer exposure to distinguish more information from a better aggregation policy.

Negative controls: include a planted wrong common premise, duplicated observations presented as multiple endorsements, a correct minority supported by a decisive test, equally acceptable alternatives, and cases with no available distinguishing evidence. Measure correctness and unsupported claims separately from agreement, decision time, cost, false blocks, and preserved constraints; calibrate thresholds only on development tasks and test on held-out tasks.

Priority: start with source-aware commitment and observable prerequisite handling, because the developmental failures concern evidence scope and missing reference loads. Keep cross-inhibition and recruitment as rival hypotheses; no quorum threshold, swarm size, or consensus rule has yet been validated for this setup.

5.4 Developmental swarm execution

Setup: a subsequent disposable Python task used two fresh investigators, one Claude Sonnet 4.6 and one GPT-6.1 Sol (medium), with separate sessions and no peer conclusions supplied during their investigations. Both could inspect the same code and four unit tests; three explicitly synthetic endorsements copied one unavailable historical DNS diagnosis from an older staging version. The current defect was a local milliseconds-to-seconds conversion dividing by 100 instead of 1000, with no network dependency in that function.

Prerequisite failure: the required evidence-and-recovery reference existed in the repository but was unavailable at the shared skill path. One investigator stopped before inspecting the fixture; the other continued and produced a diagnosis despite the missing prerequisite. Supplying the verified repository path enabled assisted resumption; this did not retroactively turn the first pass into compliant execution or repair the shared installation.

Deployment distinction: the shared skill entry and an older supporting reference were symlinks to repository files, but the new supporting reference had no shared-path entry. Updating linked source text therefore did not

install the new dependency at the path that the text required. This is an observed delivery gap, not evidence that the reference's reasoning method failed.

Observed result: after assistance, both investigators separately reproduced two failures and two passes and proposed the same divisor correction. Primary execution of a separate candidate changed only that divisor and passed all four byte-identical tests, preserving the original fixture; zero and negative-input controls passed before and after. These checks establish the exercised local behavior, not production repair, all-input correctness, or statistical independence between workers using the same code and tests.

Votes versus evidence: counting the three copied endorsements against the two current investigator recommendations gives a deliberately constructed 3:2 vote for DNS changes. This is an arithmetic comparator, not an executed naive-voting agent baseline or three real dissenting agents. Source-aware promotion instead relied on current discriminating tests and source currency; neither deduplicated head count nor consensus established truth.

Targeted critic: a fresh third diagnostic worker reran the original and candidate suites, again observing two failures versus four passes, and checked that assertions were unchanged and already-correct controls survived. It confirmed the bounded fix rather than manufacturing opposition, and challenged any inference of swarm benefit. No further diagnostic worker was recruited: this exercised a capped verification step, not an adaptive-recruitment algorithm or a validated quorum threshold.

Limits: either investigator, once unblocked, already found the correct local diagnosis; the task therefore supplies no demonstrated accuracy advantage for the swarm. The stale endorsements were transparently labeled, the defect was simple, and there was no randomized, repeated, equal-budget comparison or held-out task. Adaptive recruitment and quorum thresholds remain unvalidated; prerequisite-path delivery and stopping behavior were concrete setup failures worth addressing before expanding the team.

6 Conclusion

The chemistry supplies a vocabulary for feedback, selectivity, and sequestration—not a proof that opposing agents eliminate hallucinations. Prefer controls that preserve source lineage, distinguish observations from interpretations, and enforce permissions outside the model; test whether the metaphor-inspired additions outperform simpler, compute-matched alternatives.

Evidence status: historical and chemical facts require the sources below; the AI papers provide related precedents, not validation of this combined design. The preliminary software trials reported here are developmental observations, not a controlled benchmark or a chemistry experiment; neither effectiveness nor novelty is established.

7 References

[1] D. G. Blackmond, "The Origin of Biological Homochirality," *Cold Spring Harbor Perspectives in Biology*, vol. 2, no. 5, p. a002147, 2010, doi: 10.1101/cshperspect.a002147.

[2] The Royal Swedish Academy of Sciences, "Press release: Nobel Prize in Chemistry 2026." [Online]. Available: https://www.nobelprize.org/prizes/chemistry/2026/press-release/

[3] The Royal Swedish Academy of Sciences, "Press release: Nobel Prize in Chemistry 2021." [Online]. Available: https://www.nobelprize.org/prizes/chemistry/2021/press-release/

[4] The Royal Swedish Academy of Sciences, "They solved chemistry's asymmetric mystery." [Online]. Available: https://www.nobelprize.org/prizes/chemistry/2026/popular-information/

[5] K. Soai, T. Shibata, H. Morioka, and K. Choji, "Asymmetric autocatalysis and amplification of enantiomeric excess of a chiral molecule," *Nature*, vol. 378, pp. 767–768, 1995, doi: 10.1038/378767a0.

[6] S. V. Athavale, A. Simon, K. N. Houk, and S. E. Denmark, "Demystifying the asymmetry-amplifying, autocatalytic behavior of the Soai reaction through structural, mechanistic, and computational studies," *Nature Chemistry*, vol. 12, pp. 412–423, 2020, doi: 10.1038/s41557-020-0421-8.

[7] The Royal Swedish Academy of Sciences, "Their tools revolutionised the construction of molecules." [Online]. Available: https://www.nobelprize.org/prizes/chemistry/2021/popular-information/

[8] Y. Du and others, "Improving Factuality and Reasoning in Language Models through Multiagent Debate." [Online]. Available: https://arxiv.org/abs/2305.14325

[9] J. Huang and others, "Large Language Models Cannot Self-Correct Reasoning Yet." [Online]. Available: https://arxiv.org/abs/2310.01798

[10] K. Greshake and others, "Not what you've signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection." [Online]. Available: https://arxiv.org/abs/2302.12173

[11] S. Yao and others, "ReAct: Synergizing Reasoning and Acting in Language Models." [Online]. Available: https://arxiv.org/abs/2210.03629

[12] N. Shinn and others, "Reflexion: Language Agents with Verbal Reinforcement Learning." [Online]. Available: https://arxiv.org/abs/2303.11366

[13] JSON Schema, "Understanding JSON Schema: Object." [Online]. Available: https://json-schema.org/understanding-json-schema/reference/object

[14] S. Yao and others, "Tree of Thoughts: Deliberate Problem Solving with Large Language Models." [Online]. Available: https://arxiv.org/abs/2305.10601

[15] D. J. T. Sumpter and S. C. Pratt, "Quorum responses and consensus decision making," *Philosophical Transactions of the Royal Society B: Biological Sciences*, vol. 364, no. 1518, pp. 743–753, 2009, doi: 10.1098/rstb.2008.0204.

[16] T. D. Seeley, P. K. Visscher, T. Schlegel, P. M. Hogan, N. R. Franks, and J. A. R. Marshall, "Stop Signals Provide Cross Inhibition in Collective Decision-Making by Honeybee Swarms," *Science*, vol. 335, no. 6064, pp. 108–111, 2012, doi: 10.1126/science.1210361.

[17] F. Dietrich and K. Spiekermann, "Jury Theorems," in *The Stanford Encyclopedia of Philosophy*, Spring 2023 ed., E. N. Zalta and U. Nodelman, Eds., Metaphysics Research Lab, Stanford University, 2023. [Online]. Available: https://plato.stanford.edu/archives/spr2023/entries/jury-theorems/